Instructions for writing a literature survey (IDS)

- **Method:** Always describe the paper's methodology in the **past tense**. Example: “The authors used a Random Forest classifier…”, “Patel trained an LSTM on flow features…”.
- **Limitation:** Always state the study’s limitation in the **present tense**. Example: “The method is limited by high false positives.”, “The model is limited by high memory usage.”
- **Author citation style:**
    - **One author:** Use surname and year: *Patel (2023) found...* or *Patel (2023) proposed...*
    - **Two authors:** Use both surnames: *Sharma and Singh (2024) demonstrated...*
    - **Three authors:** Use all three surnames on first mention: *Kumar, Reddy, and Das (2022) developed...* After the first mention you may use *Kumar et al. (2022)*.
- **Structure of each paragraph (recommended):**

1. One-line context / problem statement (past tense): what they addressed.
2. One or two lines on data and features (past tense): datasets, features, preprocessing.
3. One or two lines on methods/model (past tense): classifiers, architecture, training strategy, evaluation metrics.
4. One line on results (past tense): main quantitative outcome (accuracy, F1, AUC).
5. One line limitation (present tense): single-sentence, crisp statement of what remains limited.

- **Table:** After the paragraph entries include a comparative table summarizing author, year, classifiers used, key contribution, and study limitation.

Sample literature-survey entries (detailed — three examples)

Single-author example

Patel (2023) addressed class imbalance and high-dimensional feature spaces in network intrusion detection. He used a stacked autoencoder to reduce dimensionality and then trained an SVM on the reduced feature set. Patel preprocessed the NSL-KDD dataset by removing redundant records, normalized continuous features, and applied the autoencoder for unsupervised feature learning. He trained the SVM with an RBF kernel and validated performance using 10-fold cross validation, reporting improved detection rate and reduced false positives compared to a baseline SVM trained on raw features. The approach is limited by high computational cost during real-time deployment.

(Notes: method in past tense; limitation in present tense.)

Two-author example

Sharma and Singh (2024) proposed an ensemble intrusion-detection framework that combined Random Forest and Gradient Boosting classifiers with a stacking meta-learner. They extracted flow-level and packet-level features from the CIC-IDS2018 dataset, applied SMOTE to balance minority attack classes, and trained the two base learners independently before training a logistic-regression meta-classifier on base predictions. Sharma and Singh evaluated the ensemble on labeled test splits and

reported higher overall accuracy and improved recall for rare attack classes compared to single-model baselines. The study is limited as it does not address robustness against adversarially crafted attack traffic.

(Notes: method in past tense; limitation in present tense.)

Three-author example

Kumar, Reddy, and Das (2022) developed a GAN-augmented anomaly detection pipeline to tackle extreme class imbalance in network intrusion datasets. They constructed a conditional GAN to synthesize realistic malicious flow records, used the synthetic samples to rebalance training data, and then trained a CNN classifier on the augmented dataset. The authors performed ablation studies comparing CNNs trained with and without GAN augmentation on an imbalanced benchmark and reported notable gains in detection rates for under-represented attack types. However, the approach is limited by instability during GAN training, which affects reproducibility and the quality of generated samples.

(Notes: method in past tense; limitation in present tense.)

Summary table — Sample IDS papers (2022–2025)

Author(s)	Year	Classifiers Used	Key Contribution (short)	Study Limitation (present tense)
Patel	2023	Autoencoder + SVM	Hybrid dimensionality-reduction (autoencoder) + SVM to improve detection on NSL-KDD	High computational cost in real-time is limiting
Sharma & Singh	2024	Random Forest + Gradient Boosting (stacking)	Ensemble stacking to improve recall for rare attacks on CIC-IDS2018	Lack of adversarial robustness is limiting
Kumar, Reddy & Das	2022	GAN + CNN	GAN-based sample synthesis for class balancing + CNN classifier	GAN training instability is limiting
Ali et al.	2022	LSTM	Applied sequence modeling (LSTM) to model temporal patterns in flow sequences	Long training times and sensitivity to sequence length are limiting
Mehta & Verma	2023	Random Forest	Feature-importance analysis and lightweight RF for near-real-time detection	Scalability to very large traffic volumes is limiting
Zhang	2024	XGBoost	Gradient boosting with engineered features for higher precision	Overfitting risk on small or non-representative datasets is limiting

Author(s)	Year	Classifiers Used	Key Contribution (short)	Study Limitation (present tense)
Lee et al.	2025	CNN + RNN hybrid	Temporal-spatial deep model for packet-level and flow-level features	High GPU and memory requirements are limiting
Gupta & Nair	2023	Isolation Forest	Unsupervised anomaly detection tailored for resource-constrained IoT devices	Detection of novel zero-day attacks is limited
Rahman	2024	Deep Belief Network (DBN)	Hierarchical feature learning for multi-class intrusion classification	Lack of interpretability for learned features is limiting
Chen, Wang & Liu	2025	Transformer based	Attention mechanisms applied to packet sequences for better context modeling	High memory requirements for large packet streams are limiting